

Visible-Light-Driven Photocatalytic CO₂ Reduction by a Ni(II) Complex Bearing a Bioinspired Tetradentate Ligand for Selective CO ProductionDachao Hong,^{†,‡} Yuto Tsukakoshi,[†] Hiroaki Kotani,[†] Tomoya Ishizuka,[†] and Takahiko Kojima^{*,†}[†]Department of Chemistry, Faculty of Pure and Applied Sciences, University of Tsukuba and CREST (JST), 1-1-1 Tennoudai, Tsukuba, Ibaraki 305-8571, Japan[‡]Interdisciplinary Research Center for Catalytic Chemistry, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Ibaraki 305-8565, Japan

Supporting Information

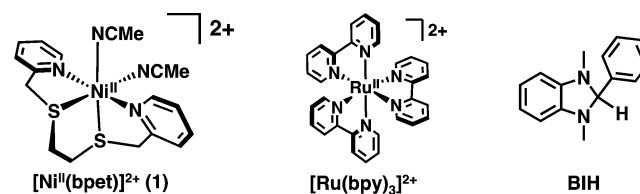
ABSTRACT: A Ni(II) complex bearing an S₂N₂-type tetradentate ligand inspired by the active site of carbon monoxide dehydrogenase was found to selectively catalyze CO₂ reduction to produce CO in a photocatalytic system using [Ru(bpy)₃]²⁺ (bpy = 2,2'-bipyridine) as a photosensitizer and 1,3-dimethyl-2-phenyl-2,3-dihydro-1H-benzo[d]imidazole (BIH) as an electron donor. The Ni(II) complex shows a high turnover number over 700 with high CO selectivity of >99% and quantum yield of 1.42% in the photocatalytic system.

Conversion of CO₂ by solar energy into chemical resources such as CO, formic acid, and methanol is a potential strategy to solve current environmental and energy problems and realize a sustainable society.^{1,2} A bottleneck is to develop a catalyst for photocatalytic CO₂ reduction because CO₂ is one of the most stable molecules and enormous energy is needed to reduce it.^{3,4} Much research has been devoted to exploring efficient catalysts for photocatalytic CO₂ reduction using both molecular metal complexes^{5–8} and heterogeneous metal nanoparticles.^{9,10} Homogeneous metal complexes provide advantages in mechanistic analysis and ligand tuning in investigations into catalysts for CO₂ reduction. So far, noble metal complexes using Re and Ru have been often used to reduce CO₂ effectively to produce CO and formate in the presence of an electron donor and a photosensitizer.^{11,12} However, developing efficient catalysts based on Earth-abundant metals remains a challenging issue in photocatalytic CO₂ reduction.

Recently, molecular catalysts based on first-row transition metals including Mn,¹³ Fe,¹⁴ Co,^{15,16} and Ni¹⁷ have been studied for photocatalytic CO₂ reduction. For example, [Mn(bpy)CO₃]Br can reduce CO₂ to produce formate using [Ru(bpy)₃]²⁺ as photosensitizer with a turnover number (TON) of 157.¹³ Photocatalytic CO₂ reduction to CO was achieved by [Fe(dmp)₂(NCS)₂] (dmp = 2,9-dimethyl-1,10-phenanthroline) with TON = 54.5.¹⁴ A macrocyclic Co complex has also reported to selectively produce CO with TON = 270 in the photocatalytic system by using a photosensitizer of [Ir(ppy)₃] (ppy = 2-phenylpyridine) that has higher reducing power for its excited state (−1.7 V vs

SCE)¹⁸ than that of one-electron-reduced [Ru(bpy)₃]²⁺ (−1.3 V vs SCE).¹⁹ A high TON of 98 000 for selective CO production has been reported when a Ni complex bearing an N-heterocyclic carbene-amine ligand was used as a catalyst under conditions including 2 nM catalyst concentration with use of [Ir(ppy)₃];¹⁷ despite the high TON, the quantum yield (QY) reached only 0.01% in that photocatalytic system.¹⁷ As for previously reported catalysts, most metal complexes bear supporting ligands with N atoms as coordination sites, which should be labile in the low-valent states of metal complexes. Thus, molecular catalysts with a new type of ligand are essential for the development of efficient photocatalytic CO₂ reduction, although large progress has been gained in the previous works. In nature, on the other hand, carbon monoxide dehydrogenase (CODH) has been identified as a CO₂ fixation enzyme, in which a Ni center is known to be ligated by sulfur atoms in the active site.²⁰ In this context, first-row metal complexes bearing multidentate ligands containing thioether–sulfur atoms together with pyridine nitrogen atoms as metal-binding sites can be expected to improve activity for CO₂ reduction in terms of stabilization of both resting and reduced species formed in the catalytic cycle. Herein, we report a Ni complex bearing an S₂N₂-type tetradentate ligand, [Ni(bpet)]²⁺ (**1**)²¹ (bpet = bis(2-pyridylmethyl)-1,2-ethanedithiol), acting as a catalyst for visible-light-driven photocatalytic CO₂ reduction. The complex allowed us to develop selective production of CO under the conditions using [Ru(bpy)₃]²⁺ as a photosensitizer and BIH as a sacrificial electron donor (Chart 1) in a dimethylacetamide (DMA)/H₂O (9:1 v/v) mixed solvent.

Chart 1. Structures of the Catalyst, Photosensitizer and Electron Donor



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Synthesis of **1** is described in the [Supporting Information \(SI\)](#), and the characterization was made by ^1H NMR and UV–vis absorption spectroscopies, mass spectrometry (MS), X-ray crystallography, and elemental analysis ([Figures S1–S4](#)). Photocatalytic CO_2 reduction was performed in a glass reactor containing DMA/ H_2O (4.0 mL, 9:1 v/v), **1** (30 μM), $[\text{Ru}(\text{bpy})_3]^{2+}$ (0.50 mM), and BIH (0.10 M). The time courses of observed products are shown in [Figure 1](#). A large

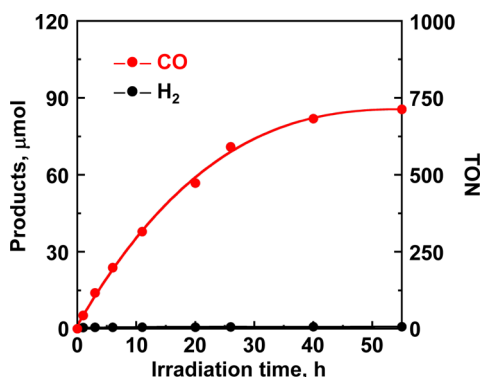


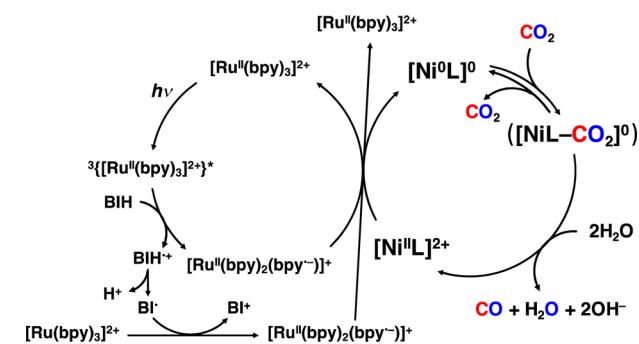
Figure 1. Time courses of products obtained from the photocatalytic reaction by irradiation (450 nm) of a DMA/ H_2O solution (4.0 mL, 9:1 v/v) containing **1** (30 μM), $[\text{Ru}(\text{bpy})_3]^{2+}$ (0.50 mM), and BIH (0.10 M) under CO_2 atmosphere at 298 K.

amount of CO (86 μmol) was evolved with a negligible amount of H_2 (0.83 μmol) at 55 h in the photocatalytic reaction. No formate was detected by GC, HPLC analysis, and ^1H NMR spectroscopy. The TON of CO production with **1** was reached 713 with CO selectivity of >99% after 55 h irradiation. The result indicates that **1** catalyzes CO_2 reduction to CO selectively. CO evolution was linearly increased from the initial irradiation ([Figure S5](#)) and saturated after 40 h, suggesting the robust catalysis of **1**. The QY of CO evolution by **1** was determined to be 1.42% ([Figure S6](#)), which is much higher than that of the reported Ni complex (0.01%),¹⁷ by the actinometer method (see [SI](#) for details and [Figure S7](#)). CO evolution was also observed under other conditions ([Table S1](#)). In addition, we examined CO_2 reduction using $^{13}\text{CO}_2$ to confirm that the origin of CO should be CO_2 ; GC-MS analysis of the produced CO clearly indicated the formation of ^{13}CO ([Figure S8](#)).

Complex **1** with S atoms coordinated to Ni center can stabilize a low-valent Ni species considered as an intermediate in the photocatalytic CO_2 reduction. Highly catalytic activity for CO_2 reduction to CO obtained with **1** implies important roles of S-coordination to the Ni center in the CODH in nature. Additionally, among photocatalytic systems using $[\text{Ru}(\text{bpy})_3]^{2+}$, macrocyclic Ni complexes such as $[\text{Ni}(\text{cyclam})]^{2+}$ have been reported to reduce CO_2 to CO; however, the TON was only 0.1, and H_2 was the main product.²² To date, there are no reports on molecular Ni complexes as efficient catalysts for selective CO production in photocatalytic systems using $[\text{Ru}(\text{bpy})_3]^{2+}$ as a photosensitizer, except for electrochemical systems.²³ The high TON and QY obtained with **1** in this work also demonstrate that **1** can serve as a functional model of CODH as well as a highly efficient catalyst in a photocatalytic system for the first time.

The photocatalytic cycle is proposed as shown in [Scheme 1](#). The reaction is initiated by photoexcitation of $[\text{Ru}(\text{bpy})_3]^{2+}$, forming the triplet MLCT excited state, $^3[\text{Ru}(\text{bpy})_3]^{2+*}$, which is reductively quenched by BIH to afford $[\text{Ru}(\text{bpy})_2(\text{bpy}^-)]^+$

Scheme 1. Proposed Mechanism of Photocatalytic CO_2 Reduction to CO Catalyzed by **1**



($\text{bpy}^{\bullet-}$)⁺ and oxidized BIH ($\text{BIH}^{\bullet+}$) that is easily deprotonated.⁵ The deprotonated $\text{BIH}^{\bullet+}$ ($\text{BI}^{\bullet}$), known as a very strong reductant, $E^0(\text{BI}^{\bullet}/\text{BI}^+) = -2.06$ V vs SCE, can reduce another $[\text{Ru}(\text{bpy})_3]^{2+}$ to produce 2 equiv of $[\text{Ru}(\text{bpy})_2(\text{bpy}^{\bullet-})]^+$ in the overall photoinduced process.⁵ The rate constant of the reductive quenching of $^3[\text{Ru}(\text{bpy})_3]^{2+*}$ by BIH was determined to be $5.6 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$ from a Stern–Volmer plot in DMA/ H_2O (9:1 v/v) solution ([Figure S9](#)). Finally, $[\text{Ru}(\text{bpy})_2(\text{bpy}^{\bullet-})]^+$ can reduce the complex **1** to afford a $[\text{Ni}^0(\text{bpet})]^0$ ($[\text{Ni}^0\text{L}]^0$) intermediate that reacts with CO_2 to produce CO selectively.

To gain mechanistic insight into the photocatalytic CO_2 reduction by **1**, the electrochemical properties of **1** were investigated by cyclic voltammetry (CV) and differential pulse voltammetry (DPV). The CVs and DPVs of **1** in DMA/ H_2O solution under Ar and CO_2 atmosphere are shown in [Figure 2](#).

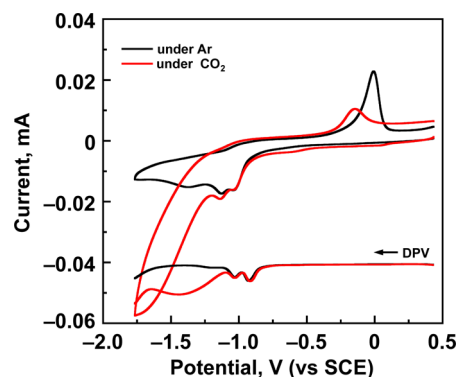


Figure 2. CVs and DPVs of **1** (1.0 mM) in a DMA/ H_2O (9:1 v/v) solution containing 0.10 M TBAPF₆ (TBA = tetra-*n*-butylammonium) under Ar or CO_2 atmosphere.

The two irreversible reductive waves of **1** observed under both Ar and CO_2 can be assigned to be the reduction of the Ni center from Ni^{II} to Ni^{I} and Ni^{I} to Ni^0 . The reduction potentials of **1** were determined to be $E_{\text{pc}}(\text{Ni}^{\text{II}}/\text{Ni}^{\text{I}}) = -0.93$ V and $E_{\text{pc}}(\text{Ni}^{\text{I}}/\text{Ni}^0) = -1.03$ V (vs SCE) from the DPVs. The small potential difference between Ni^{I} and Ni^0 could be attributed to the effect of S coordination that is expected to stabilize Ni^0 species and solvent effect ([Figure S10](#)). The irreversible reduction waves can be explained by structural changes of **1** from octahedral Ni^{II} to tetrahedral Ni^0 complexes in the d^{10} electronic configuration. The peak around -0.2 V is attributed to the irreversible oxidation wave on the basis of the scan-rate dependence ([Figure S11](#)). The catalytic current was observed only under CO_2 after the second reduction wave, suggesting

CO₂ was reacted with Ni⁰ species to generate CO in the electrochemical process. The onset potential of catalytic current observed at −1.2 V (vs SCE) indicates that photochemically formed [Ru(bpy)₂(bpy^{•−})]⁺ ($E_{\text{ox}} = -1.3$ V vs SCE)¹⁹ can potentially reduce **1** to afford [Ni⁰L]⁰ as the reactive species.

Concentration dependence of **1** on CO₂ reduction activity was also studied under the same conditions, as shown in Figure 3. CO evolution was increased in proportion to the increase of

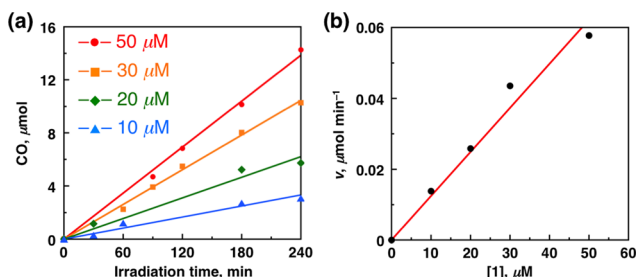


Figure 3. (a) Time courses of products obtained from the photocatalytic reaction by irradiation (450 nm) of a DMA/H₂O solution (4.0 mL, 9:1 v/v) containing **1** (blue triangles: 10 μM, green diamonds: 20 μM, orange squares: 30 μM, and red dots: 50 μM), [Ru(bpy)₃]²⁺ (0.50 mM), and BIH (0.10 M) under CO₂ atmosphere. (b) Concentration dependence of **1** on CO evolution rate.

concentration of **1** in Figure 3a: the initial rate of CO evolution exhibited a first-order dependence on the concentration of **1**. Concentration dependence of CO₂ exhibited saturation behavior as shown in Figure S12. The results indicate the existence of coordination equilibrium between CO₂ and the [Ni⁰L]⁰, which is involved in the rate-determining step (RDS) of the CO formation (Scheme 1). The turnover frequency (TOF) of CO evolution calculated from the slope in Figure 3b was determined to be 0.31 min⁻¹, which is comparable to the reported values of the aforementioned Fe (0.38 min⁻¹)¹⁴ and Co complexes (0.37 min⁻¹).¹⁵

The influence of water molecules on the reaction mechanism was also investigated in terms of solvent kinetic isotope effect (sKIE) and H₂O concentration dependence. The TOF in DMA/D₂O (9:1 v/v) was determined to be 0.26 min⁻¹ (Figure S13), giving an sKIE value of 1.20 ± 0.16. As described in Table S2, the higher H₂O concentration enhanced the reaction efficiency of not only CO but also H₂ evolution. However, the highest CO selectivity was achieved under the DMA/H₂O (9:1) conditions. Judging from the [H₂O] dependence, H₂O acts as the proton source and should be one of factors controlling the selectivity of CO/H₂ evolution.

Several control experiments were performed to demonstrate **1** as an efficient molecular catalyst for photocatalytic CO₂ reduction to CO, as shown in Figures 4 and S14. In the absence of **1**, a negligible amount of CO was observed in a continuous photoirradiation, indicating that **1** is required as the catalyst for CO₂ reduction. Under Ar atmosphere, almost no generation of CO and H₂ was observed in the presence of **1** after 20 h irradiation at 450 nm, supporting that CO is derived exclusively from CO₂, together with the ¹³C-labeling experiments as mentioned above. To examine the possibility of heterogeneous catalysis by nanoparticles derived from Ni ions, Ni(ClO₄)₂ was employed in the photocatalytic CO₂ reduction; only a little amount of CO was observed as in the reaction without **1**. The result allows us to exclude the possibility of CO₂ reduction by Ni nanoparticles from Ni(ClO₄)₂. Additionally, dynamic light

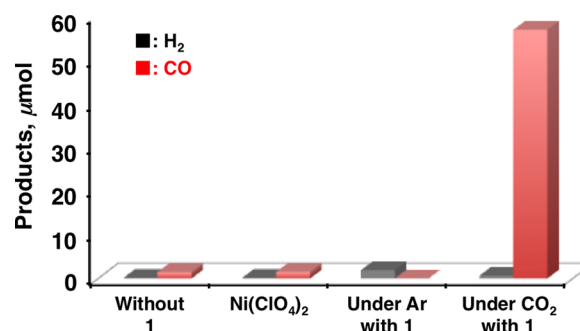


Figure 4. Products obtained from the photocatalytic reaction of irradiation (450 nm) of DMA/H₂O (9:1 v/v) solutions at 20 h containing [Ru(bpy)₃]²⁺ (0.50 mM) and BIH (0.10 M) without a catalyst, with Ni(ClO₄)₂ (30 μM), or with **1** (30 μM) under Ar and under CO₂ atmosphere at 298 K. (See Figure S10 for the time courses.)

scattering (DLS) measurements of the reaction solution including **1**, [Ru(bpy)₃]²⁺, and BIH before and after irradiation indicated no formation of nanoparticles (Figure S15). Photocatalytic CO₂ reduction of **1** with a drop of Hg also exhibited no significant difference in CO evolution (Figure S16). Judging from these control experiments together with DLS measurements, we concluded that **1** clearly catalyzes CO₂ reduction to CO in a homogeneous manner.

The electrochemical and kinetic data, together with reference experiments, also lend credence to the proposed mechanism of CO₂ reduction by **1** as shown in Scheme 1. Complex **1** ([Ni^{II}L]²⁺) is reduced to form [Ni⁰L]⁰, which subsequently reacts with CO₂ to afford the intermediate, [NiL-CO₂]⁰. The intermediate finally releases CO through the reaction of the CO₂ ligand with H₂O accompanying the O–H bond cleavage, and returns to [Ni^{II}L]²⁺. The RDS of CO₂ reduction by **1** is the CO₂ coordination to the reduced [Ni⁰L]⁰ intermediate, although further characterization is required to investigate the intermediates derived from **1**.

In summary, we have developed an efficient photocatalytic CO₂ reduction system using **1**, bearing an S₂N₂-type ligand inspired by the CODH enzyme as a homogeneous catalyst, to produce CO selectively in the presence of [Ru(bpy)₃]²⁺ as a photosensitizer and BIH as an electron donor. The TON and QY reached 713 and 1.42%, respectively, which are the highest values among reported Ni complexes in the photocatalytic CO₂ reduction systems using [Ru(bpy)₃]²⁺. Based on electrochemical and kinetic results, the RDS of CO₂ reduction can be the step of CO₂ binding to the [Ni⁰L]⁰ species. Finally, this work has expanded the possibility of ligand design for the development of efficient and selective photocatalytic CO₂ reduction using metal-complex catalysts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.7b01956.

Experimental details and analytical data, including Figures S1–S16 and Tables S1 and S2 (PDF)
X-ray crystallographic data for **1** (CIF)

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Notes

The authors declare no competing financial interest.

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